

INOWROCLAW, Poland

WMO: 123420

Lat: 52.807N

Lon: 18.277E

Elev: 83

StdP: 100.33

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-12.9	-17.9	0.8	-14.6	-15.7	1.0	-12.1	13.4	0.6	12.4	1.0	3.0	80	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.1	20.1	29.2	19.3	27.3	18.6	21.3	28.2	20.5	27.1	19.6	25.5	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.2	23.4	18.4	13.4	22.4	17.4	12.6	21.9	62.2	28.9	59.2	26.8	56.4	25.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.1	9.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-2.3	-0.9	3.8	9.7	14.2	17.9	19.8	19.4	14.3	8.6	4.8	0.9
	DBStd	8.67	5.36	5.61	4.70	4.29	4.19	3.26	3.17	3.14	3.25	4.00	3.38	4.82
	HDD10.0	1468	381	304	196	58	11	0	0	0	3	77	158	282
	HDD18.3	3496	639	538	450	261	139	48	20	24	127	303	406	540
	CDD10.0	1190	0	0	4	48	140	236	305	291	133	32	1	0
	CDD18.3	177	0	0	0	2	11	33	67	57	7	0	0	0
	CDH23.3	1761	0	0	0	31	143	351	599	547	90	0	0	0
	CDH26.7	561	0	0	0	8	26	110	204	181	32	0	0	0
(14) Wind	WSAvg	4.4	4.9	4.7	4.7	4.7	4.2	4.0	3.9	3.8	4.1	4.3	4.5	5.0
(15) Precipitation	PrecAvg	524	30	26	34	30	54	56	85	57	48	34	36	34
	PrecMax	743	73	60	65	99	134	115	170	176	113	113	124	88
	PrecMin	355	3	4	10	4	14	21	27	13	12	5	2	7
	PrecStd	87	17	12	15	19	25	27	40	34	27	24	21	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	10.0	18.9	27.9	29.1	33.8	33.7	33.6	30.5	21.7	14.0	11.2
		MCWB	7.6	7.2	11.8	16.4	17.9	20.6	20.7	21.0	19.2	15.9	12.3	9.4
	2%	DB	7.8	8.6	15.0	22.8	27.0	29.5	31.1	30.7	25.4	18.6	11.8	9.3
		MCWB	6.4	6.6	9.3	13.8	17.0	19.9	20.3	20.0	17.4	14.4	10.3	8.2
	5%	DB	6.0	7.1	12.7	20.1	24.8	27.1	29.0	28.8	22.5	16.5	10.6	8.1
		MCWB	5.1	5.3	8.6	12.5	16.2	18.5	19.5	19.4	15.7	13.3	9.3	6.9
	10%	DB	4.3	5.6	10.3	17.7	22.4	24.8	26.7	26.5	20.2	14.6	9.3	6.7
		MCWB	3.4	4.1	7.2	11.3	15.0	17.5	18.8	18.4	14.9	12.2	8.1	5.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	8.1	12.1	16.5	19.1	22.2	22.5	22.1	19.9	16.6	12.3	9.9
		MCDB	8.8	9.4	16.6	24.8	26.1	29.3	28.2	29.3	28.6	20.6	13.6	10.7
	2%	WB	6.4	6.7	10.5	14.8	18.0	20.9	21.3	21.0	18.4	14.9	10.6	8.5
		MCDB	7.3	8.2	13.9	21.9	25.2	27.8	27.8	28.6	24.3	17.6	11.6	9.3
	5%	WB	5.0	5.4	8.9	13.3	17.0	19.5	20.6	20.2	16.7	13.7	9.4	6.9
		MCDB	6.0	6.9	12.2	18.8	23.3	25.4	26.7	26.8	20.5	16.1	10.4	7.9
	10%	WB	3.4	4.2	7.5	11.9	15.9	18.3	19.7	19.3	15.5	12.4	8.2	5.6
		MCDB	4.2	5.4	9.9	16.5	21.2	23.8	25.1	24.9	19.2	14.4	9.2	6.6
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.9	8.2	10.7	10.8	10.8	10.5	10.9	9.5	7.5	5.1	4.5
		MCDBR	4.9	6.9	12.0	15.2	14.3	14.3	14.0	14.4	13.9	10.0	6.7	5.3
		MCWBR	4.2	5.3	7.5	8.0	6.1	6.3	5.3	5.8	6.6	6.3	5.4	4.7
	5% WB	MCDBR	4.8	6.5	10.7	13.5	12.7	12.3	12.0	12.4	11.6	9.1	6.5	5.2
		MCWBR	4.2	5.4	7.0	7.4	6.3	6.8	5.3	5.7	6.5	6.1	5.4	4.8
(40) Clear-Sky Solar Irradiance	taub		0.312	0.341	0.374	0.420	0.415	0.408	0.440	0.429	0.387	0.366	0.346	0.309
	taud		2.268	2.270	2.239	2.165	2.233	2.284	2.210	2.244	2.335	2.377	2.330	2.296
	Ebn at Noon		658	754	803	809	836	845	806	792	784	719	613	591
	Edh at Noon		64	86	109	133	130	125	133	122	99	78	61	53
(44) All-Sky Solar Radiation	RadAvg		0.64	1.32	2.56	4.15	5.19	5.42	5.00	4.41	3.16	1.73	0.72	0.46
	RadStd		0.11	0.21	0.34	0.54	0.54	0.49	0.59	0.39	0.42	0.26	0.09	0.06

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-169	+80	N/A

Nomenclature: See separate page